

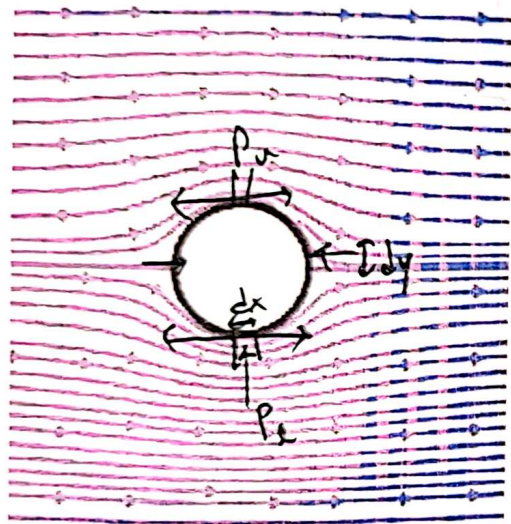
Potential flow over a spinning cylinder (TSec. 3.15)

Elementary potential flows { uniform flow
source/sink
Doublet
vortex

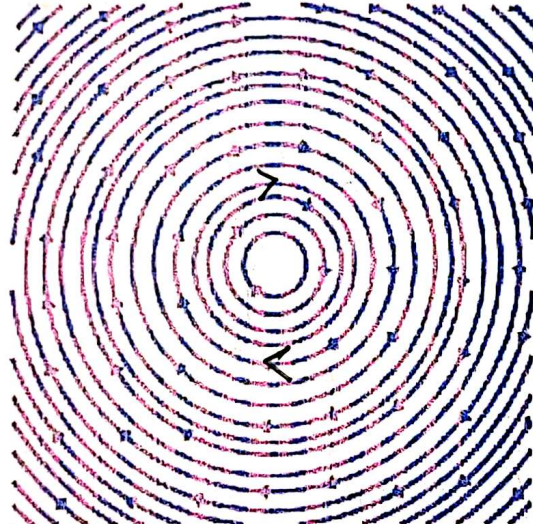
Non-spinning cylinder

Vortex

Spinning cylinder



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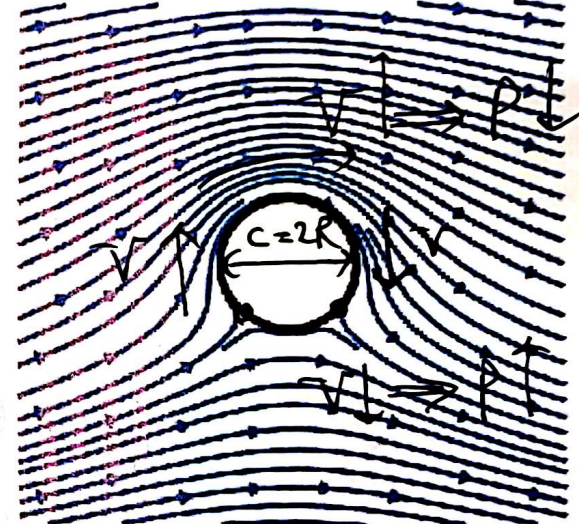


Figure 2.23: Potential flow over a spinning cylinder. Based on image from FA-JA

$$c_l = \int^c (c_{p_l} - c_{p_u}) \frac{dx}{c} = 0$$

$$c_d = \int^c (c_{p_u} - c_{p_l}) \frac{dy}{c} = 0$$

$$x = c \cos \theta$$

$$dx = -c \sin \theta d\theta$$

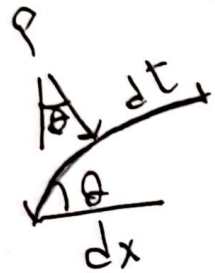
$$c_l = \int_{c=2R}^c (c_{p_l} - c_{p_u}) \frac{dx}{c} \neq 0$$

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} + \psi_{\text{vortex}} \quad (2.117)$$

$$= \psi_{\text{non-spinning cylinder}} + \psi_{\text{vortex}} \quad (2.118)$$

$$c_d = f(dy)$$

$$y = c \sin \theta \rightarrow dy = c \cos \theta d\theta$$

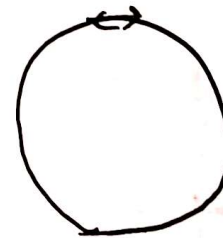


$$dt \cos \theta = \underline{\underline{dx}}$$

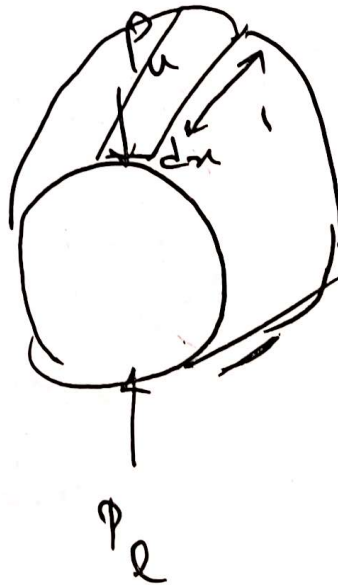
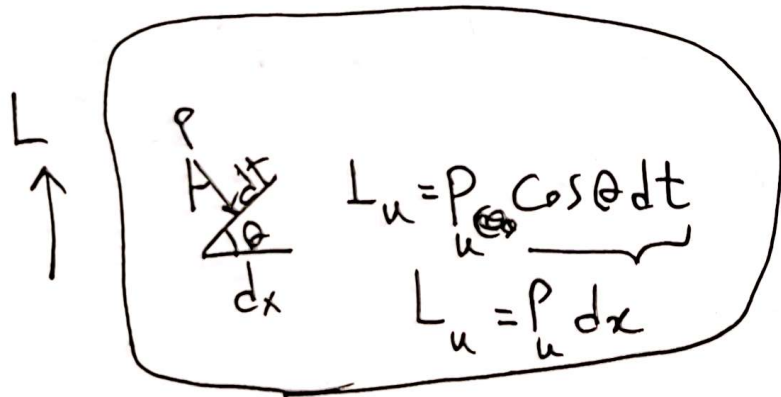
$$A = \underline{\underline{dx}} (1)$$



$$L_u = -P_u dx$$



$$L_l = P_l dx$$



$$L = L_l + L_u = \int (P_l - P_u) dx$$

$$C_l = \frac{1}{c} \int (C_{p_l} - C_{p_u}) dx$$

note that we ignored
viscosity → ideal flow

Potential flow over a spinning cylinder

$$\frac{\Gamma}{2\pi} (\ln r - \ln R) = \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

$$\psi(r, \theta) = U_{\infty} r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right] + \frac{\Gamma}{2\pi} \ln \frac{r}{R} \quad (2.119)$$

$\psi_{\text{non-spin}} + \psi_{\text{vortex}}$

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_{\infty} \cos \theta \left[1 - \frac{R^2}{r^2} \right] \rightarrow \text{same as non-lifting cyl.} \quad (2.120)$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r} = -U_{\infty} \sin \theta \left[1 + \frac{R^2}{r^2} \right] - \frac{\Gamma}{2\pi r} \quad (2.121)$$

non-lifting vortex effect

$$C_p = \underbrace{1 - 4 \sin^2 \theta}_{\text{non-lifting}} + \underbrace{\frac{-2\Gamma \sin \theta}{\pi R U_{\infty}}}_{\text{vortex-non}} - \underbrace{\left(\frac{\Gamma}{2\pi R U_{\infty}} \right)^2}_{\text{pure vortex}} \quad (2.122)$$

$$\downarrow$$

$$C_l = .$$

$$C_d = .$$

lifting inter action
gives lift

$$C_p = 1 - \left(\frac{V}{U_{\infty}} \right)^2$$

$$V = \sqrt{V_r^2 + V_{\theta}^2}$$

$$\Rightarrow V^2 = V_r^2 + V_{\theta}^2$$

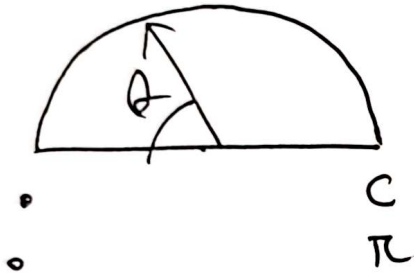
c_l & c_d are $\int$ of c_p

$$c_p \text{ is f } (V^2) \rightarrow \begin{cases} c_{p \text{ non-lifting}} : \underline{-4 \sin^2 \theta} \\ c_p : \text{vortex \& vortex-non-lifting} \\ \text{interaction} \end{cases}$$

Component
 $\rightarrow c_p$ ~~term~~ of vortex-non-lifting interaction
gives c_l which is equal

$$t_a : \frac{\Gamma}{R V_\infty}$$

Potential flow over a spinning cylinder



$$C_d = \int_0^c C_p - C_{p2} dy$$

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p (\cos \theta d\theta) = 0 \quad (2.123)$$

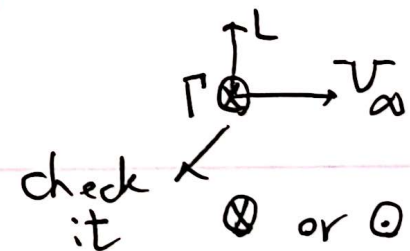
$$C_l = -\frac{1}{2} \int_0^{2\pi} C_p (\sin \theta d\theta) = \left[\frac{\Gamma}{RU_\infty} \right] \text{ after some algebra} \quad (2.124)$$

$$l = \frac{1}{2} C_l \rho_\infty U_\infty^2 c = \frac{1}{2} \frac{\Gamma}{RU_\infty} \rho_\infty U_\infty^2 2R = \left[\rho_\infty U_\infty \Gamma \right] \quad (2.125)$$

$$C_l = \frac{l}{\frac{1}{2} \rho_\infty U_\infty^2 c}$$

$$C_d = 0, C_l = \frac{\Gamma}{RU_\infty}, l = \rho_\infty U_\infty \Gamma$$

$$\vec{L} = \rho_\infty \vec{U}_\infty \times \vec{\Gamma}$$



Spinning" the cylinder causes an asymmetrical force due to Γ which in turn causes lift $\rightarrow$ Magnus effect

Potential flow over a spinning cylinder

for example $\sin^{-1} 2 = !$

- Stagnation point $\rightarrow \theta = \sin^{-1} -\Gamma / 4\pi R U_\infty$, $r = R$. (Show as an ungraded homework)
- Max velocity $\rightarrow r = R$, $\theta = \pi/2$ (Show as an ungraded homework)
- Stagnation streamline is still $\psi = 0$ (Show as an ungraded homework)
- At $r = R$, $v_r = 0$ and $v_\theta = -2U_\infty \sin \theta - \Gamma / 2\pi R$ (Show as an ungraded homework)
- $C_l = \Gamma / R U_\infty$
- $C_d = 0$ Can also determine by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.



Note: Since the spinning cylinder produces lift sometimes we refer to it as the lifting cylinder and the non-spinning cylinder as a non-lifting cylinder

otential flow over a spinning cylinder

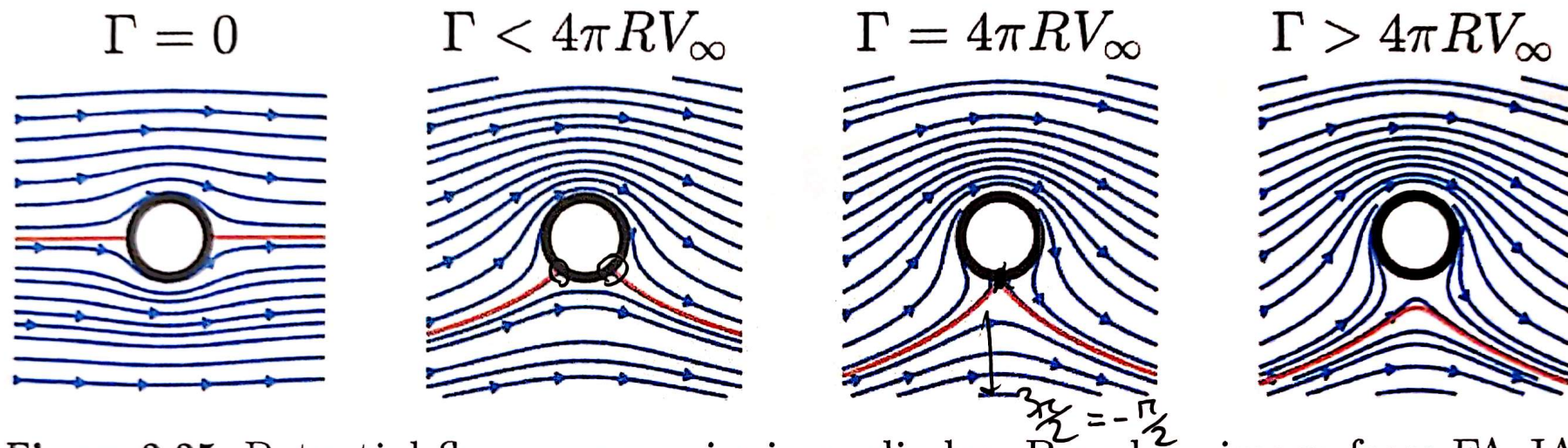


Figure 2.25: Potential flow over a spinning cylinder. Based on image from FA-JA

Γ determines the locations of the stagnation points. Γ is also directly proportional to how fast the cylinder is spinning and also to the lift.

$$L = \oint_{\gamma} V_{\infty} \Gamma \quad \text{for cyl.}$$

$$L = L_{\xi-\eta} = \oint_{\gamma} V_{\infty} \Gamma$$

different flow regimes

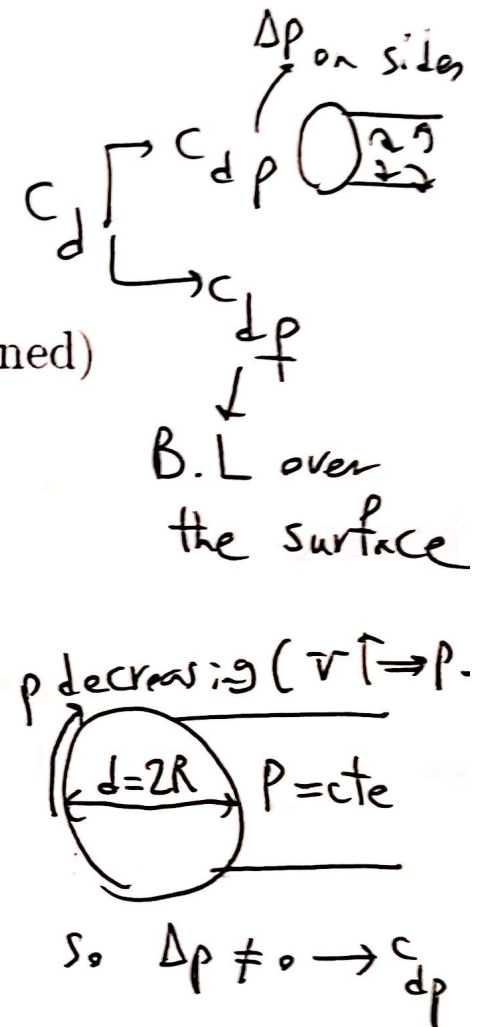
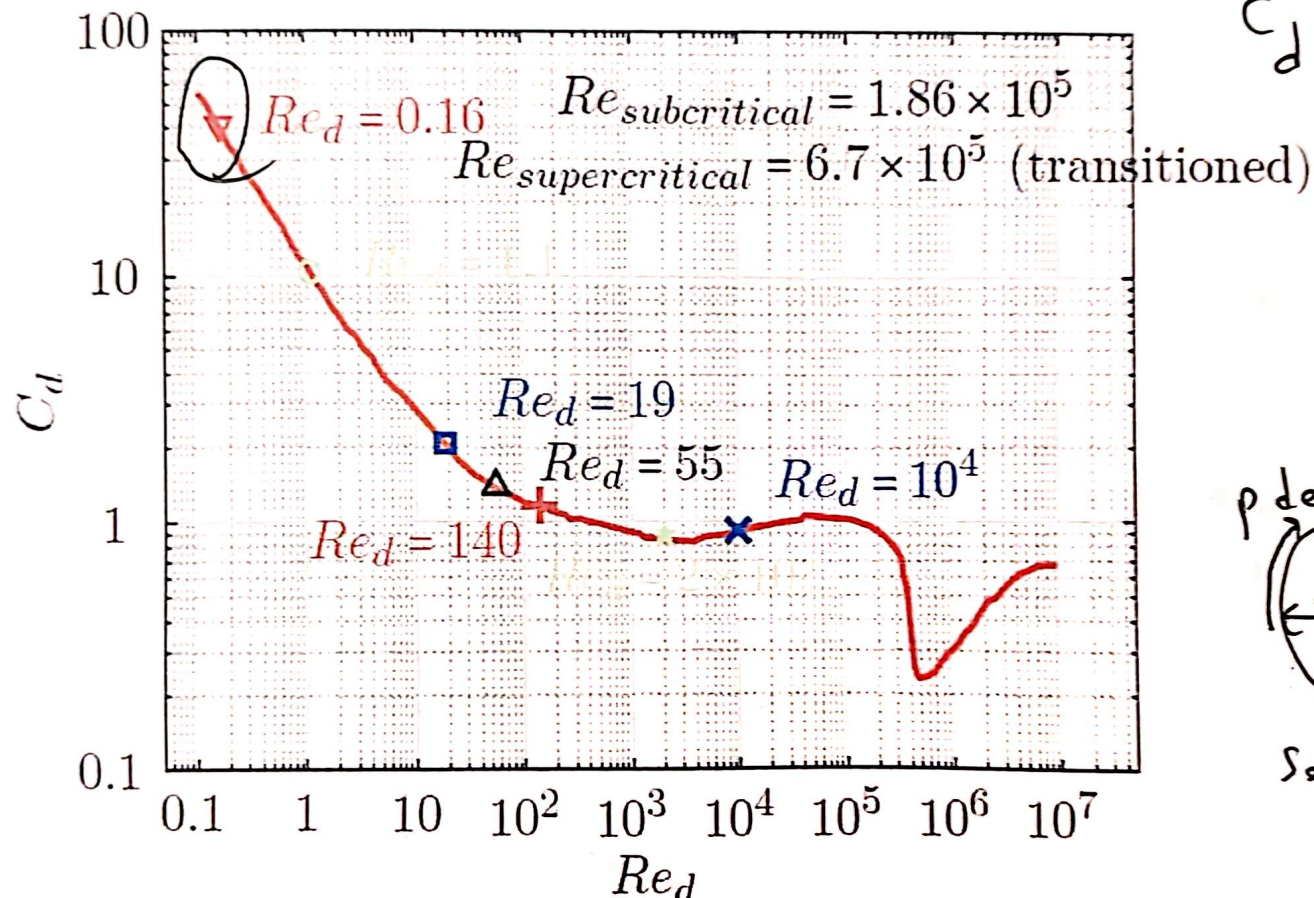


Figure 3.2: Variation of Drag Coefficient C_d of a cylinder as a function of the Reynolds number Re_d

Need to understand, boundary layers, pressure gradients and flow separation to explain.

$$\phi = -\frac{A \cos \theta}{2\pi\lambda}$$

①

(a) Study incompressible continuity equation for $z=0$

$$\frac{1}{\lambda} \left(\frac{\partial}{\partial r} (v_r r) \right) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} = 0$$

Cont. in polar coordinate.

$$\frac{v_r}{\lambda} + \frac{\partial v_r}{\partial r} + \frac{1}{\lambda} \frac{\partial v_\theta}{\partial \theta} = 0 \rightarrow \textcircled{1}$$

$$v_r = \frac{\partial \phi}{\partial r} = \frac{A \cos \theta}{2\pi\lambda^2}$$

$$v_\theta = \frac{1}{\lambda} \frac{\partial \phi}{\partial \theta} = \frac{A \sin \theta}{2\pi\lambda^2}$$

Put into $\textcircled{1}$ $\nabla^2 \phi = \textcircled{2}$

$$\frac{A \cos \theta}{2\pi\lambda^3} + \frac{A \cos \theta}{2\pi\lambda} \left(\frac{-1}{\lambda^3} \right) + \frac{1}{\lambda} \frac{A \cos \theta}{2\pi\lambda^2} = 0$$

$\Rightarrow$ Satisfied $\therefore$ flow is possible

Also can use $\nabla^2 \phi = 0$ or $\nabla^2 \psi = 0$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

$$\frac{\partial}{\partial r} \left[\frac{+A \cos \theta}{2\pi\lambda^2} \right] + \frac{1}{\lambda} \left[\frac{A \cos \theta}{2\pi\lambda^2} \right] + \frac{1}{\lambda^2} \frac{\partial}{\partial \theta} \left[\frac{A \sin \theta}{2\pi\lambda} \right]$$

$$\Rightarrow \frac{-2A}{\pi\lambda^3} \cos \theta + \frac{A \cos \theta}{2\pi\lambda^3} + \frac{1}{\lambda^2} \frac{A (\cos \theta)}{2\pi\lambda} = 0$$

$\Rightarrow$ also satisfied